

Report 02

Problem Statement Analysis & PRD Formulation

Project:	JaJeev (formerly ORCA) — Agentic Marine Intelligence System
Phase:	Requirements Engineering
Period:	August 2026 (Week 1–2)
SIH Problem Statement:	SIH26176 (ISRO)

1. Executive Summary

Following the field research documented in Report 01, Team ORCA formalised the findings into a comprehensive Product Requirements Document (PRD) — a 3,000+ line specification that became the single source of truth for the entire project. This report documents the analytical process of translating field observations into engineering requirements, the problem decomposition strategy, and the key architectural decisions made before any code was written.

2. ISRO's Problem Statement — Decoded

2.1 What ISRO Explicitly Requires (SIH26176)

1. Natural language query interface for diverse marine stakeholders
2. Automatic intent identification and task decomposition
3. Multi-agent autonomous coordination
4. Integration of satellite EO, GIS, weather, oceanographic, and advisory data
5. Spatial, temporal, and contextual reasoning across heterogeneous sources
6. Explainable, evidence-based recommendations via conversational interface
7. Indian regional language support including voice interaction
8. Proactive alerts: adverse weather, cyclones, lightning, high waves
9. Geofencing: international boundaries, restricted waters, ecologically sensitive zones
10. Route optimisation and safe navigation assistance

2.2 Representative Queries ISRO Expects the System to Handle

- *"Where is the nearest Potential Fishing Zone (PFZ) today?"*
- *"Is it safe to venture into the sea tomorrow morning?"*
- *"What are the tide, weather, and sea conditions near my fishing location?"*

- *"Are there any lightning or cyclone alerts in my area?"*
- *"Which regions show high chlorophyll and favourable SST?"*
- *"What is the safest route for a fishing vessel considering weather and sea-state?"*
- *"Why has fish productivity declined in a particular coastal region?"*
- *"Which fishing zones should be avoided due to hazardous conditions or geofencing restrictions?"*

2.3 What ISRO Implicitly Expects

Beyond the explicit requirements, ISRO's use of the word **"agentic"** and their mention of **satellite EO integration** told us:

- This is NOT a chatbot project — ISRO wants real data flowing through real agents
- ISRO already has the data (MOSDAC, INCOIS) — they want the **intelligence layer** on top
- Explainability is not optional — ISRO evaluates scientific rigour, not demo polish
- Deployment-readiness matters — "could ISRO actually deploy this?"

3. Solution Positioning

3.1 The One-Line Pitch

"JaJeev is an Agentic Marine Intelligence System that converts heterogeneous Indian ocean observations into explainable operational decisions for fishermen, coastal authorities, and maritime operators — in their own language."

3.2 Differentiation from Typical Hackathon Solutions

Typical Hackathon Solution	JaJeev
One LLM + one weather API	Multi-agent planner over 15+ live data sources
Generic map overlay	H3-indexed Marine Digital Twin with per-cell marine state vector
English only	10 Indian regional languages + voice I/O
Online only	Full offline mode with service workers + cached advisories
Single static recommendation	Risk-ranked candidates + evidence graph + rejected alternatives
No confidence score	Uncertainty engine — per-variable confidence + data freshness
LLM computes everything	LLM explains only — all computation is deterministic Python + ML

4. Core Design Philosophy Established

The PRD enshrined five non-negotiable principles that govern every subsequent engineering decision:

Principle 1: The Processing Pipeline

OBSERVE → UNDERSTAND → REASON → PREDICT → SIMULATE → DECIDE → EXPLAIN → ACT

Every component maps onto this chain. No step is skipped. Every output is auditable.

Principle 2: LLM Never Computes

The LLM's only two jobs: (1) parse natural-language intent into structured parameters, (2) phrase a final answer from an evidence receipt. If an LLM is ever asked to estimate a physical value, **that is a bug**.

Principle 3: LLM is Optional

Both LLM jobs must have a tested, non-degraded fallback. The system must work with zero LLM calls.

Principle 4: Never Fabricate

Every adapter that can't verify its data returns {"status": "unavailable"} rather than a guess. Real labels vs. synthetic labels are distinguished loudly.

Principle 5: Hard Constraints Veto; Soft Constraints Only Rank

An MPA violation cannot be outweighed by calm seas. A certain-capsize wave height is a veto, not a score contribution.

5. Data Source Inventory & Registration Plan

The PRD catalogued every data source across three tiers:

Tier 1 — Core Operational Data

Source	Data	Registration
INCOIS OSF (ERDDAP)	Wave, current, SST, MLD, tides	No registration (public)
INCOIS PFZ Archive	Fishing zone training labels (2003–2025)	Portal access needed
ISRO MOSDAC	EOS-06 chlorophyll, satellite products	Free registration (1–2 days)
IMD	Cyclone tracks, lightning, weather forecast	Free API key (1–3 days)
Copernicus Marine	Physics, waves, biogeochemistry	Free account (same day)
ERA5/CDS	Historical atmospheric reanalysis	Free account (same day)
NOAA OISST	Global daily SST since 1981	Public ERDDAP (no auth)

Tier 2 — Geospatial Boundaries

Source	Data
Marine Regions	World EEZ v12 (200nm boundaries)
Protected Planet	Marine Protected Areas
GEBCO 2026	Global bathymetry (15 arc-second)

Tier 3 — Differentiators

Source	Data
Global Fishing Watch	Historical fishing effort (2012–present)
Open-Meteo Marine	Third independent wave/wind source
BGC-Argo	Subsurface biogeochemistry profiles

Day 1 Action: Register for MOSDAC, IMD, Copernicus Marine, CDS, NASA Earthdata, Protected Planet, GFW — all free, some take 2–5 days to activate.

6. User Story Specification

The PRD defined 15+ user stories across stakeholder groups. Key examples:

ID	As a fisherman I want to...	Priority
US-F01	Ask in Odia whether it's safe to fish tomorrow	P0
US-F02	See the nearest PFZ zone on a map	P0
US-F03	Know wave height at my planned zone	P0
US-F04	Get a phone alert if a cyclone is approaching	P0
US-F05	Know if I'm approaching a restricted area	P0
US-F08	See the safest route to my fishing zone	P1
US-F09	Ask "what if I leave at 8 AM instead?"	P2

7. Multi-Source Redundancy Matrix

A critical design decision: **no single point of failure for any variable.**

Variable	Source 1	Source 2	Source 3	Consensus
SST	INCOIS	MOSDAC OCM-3	NOAA OISST	Weighted mean; flag if >1°C spread
Wave height	INCOIS OSF	Copernicus Marine	Open-Meteo Marine	Ensemble mean; report std dev
Ocean current	INCOIS	Copernicus Marine	HYCOM	Ensemble mean; flag if >0.3 m/s spread
Wind speed	IMD	ERA5	Open-Meteo	Ensemble mean; IMD highest priority
Chlorophyll	MOSDAC OCM-3	NASA OceanColor	Copernicus BGC	Weighted by cloud-free coverage

8. Phased Roadmap Established

Phase	Period	Scope
Phase 0	Aug 25 – Sep 20	DB + adapters + core agents + React PWA + internal round submission
Phase 1	October	Fishing Agent + ML model + full orchestration
Phase 2	November	Simulation + anomaly detection + route optimisation + languages
Phase 3	Late Nov – Dec	Offline mode + voice I/O + proactive alerts + demo prep

9. PRD Deliverable

The final PRD (ORCA_PRD_SIH26176_v2.md) was delivered at **3,055 lines** covering 35 sections including:

- Full system architecture diagrams
- All 11 agent specifications with inputs, outputs, and tools
- Complete database schema (PostGIS, TimescaleDB, pgvector)
- API specification for every endpoint
- ML model specification (XGBoost Fishing Suitability)
- Testing strategy (unit, integration, backtest)
- Risk assessment with mitigations
- Demo script with specific coordinate points

10. Key Decision: Naming

The project was originally named **ORCA** (Marine EcOsystem Reasoning with Collaborative Agents). During this phase it was renamed to **JalJeev** (जलजीव — "Life of Water" in Hindi/Sanskrit) to better reflect its Indian identity and purpose. The PRD filename retains the ORCA name for continuity.

*Report prepared by Team ORCA, KIIT University, Bhubaneswar
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